

ESP GLASSES SCANNER

ANNOTATED PHOTO BOOK

CYD / CYD Expanded / ESP32-S3 / ESP32-WROOM



How to use this photo book

Each photograph is reproduced at large size with a short technical annotation panel underneath.

The original labels embedded in the supplied photographs have been preserved. The added notes standardise terminology and explain what each image is intended to demonstrate.

The sequence moves from CYD interface states, through LED states, to hardware identification and comparison.

LED quick reference

	BLUE	Boot / startup
	GREEN	Ready / normal scanning
	ORANGE	Warning / possible detection
	RED	High-priority / high-confidence alert

Photo sequence

- 1-5: CYD operating and startup interfaces
- 6-10: ESP32-S3 / ESP32-WROOM LED states and comparisons
- 11-12: CYD sentry and legal/privacy pages
- 13: final board-to-board hardware comparison

01. CYD / CYD Expanded - Live scanning page



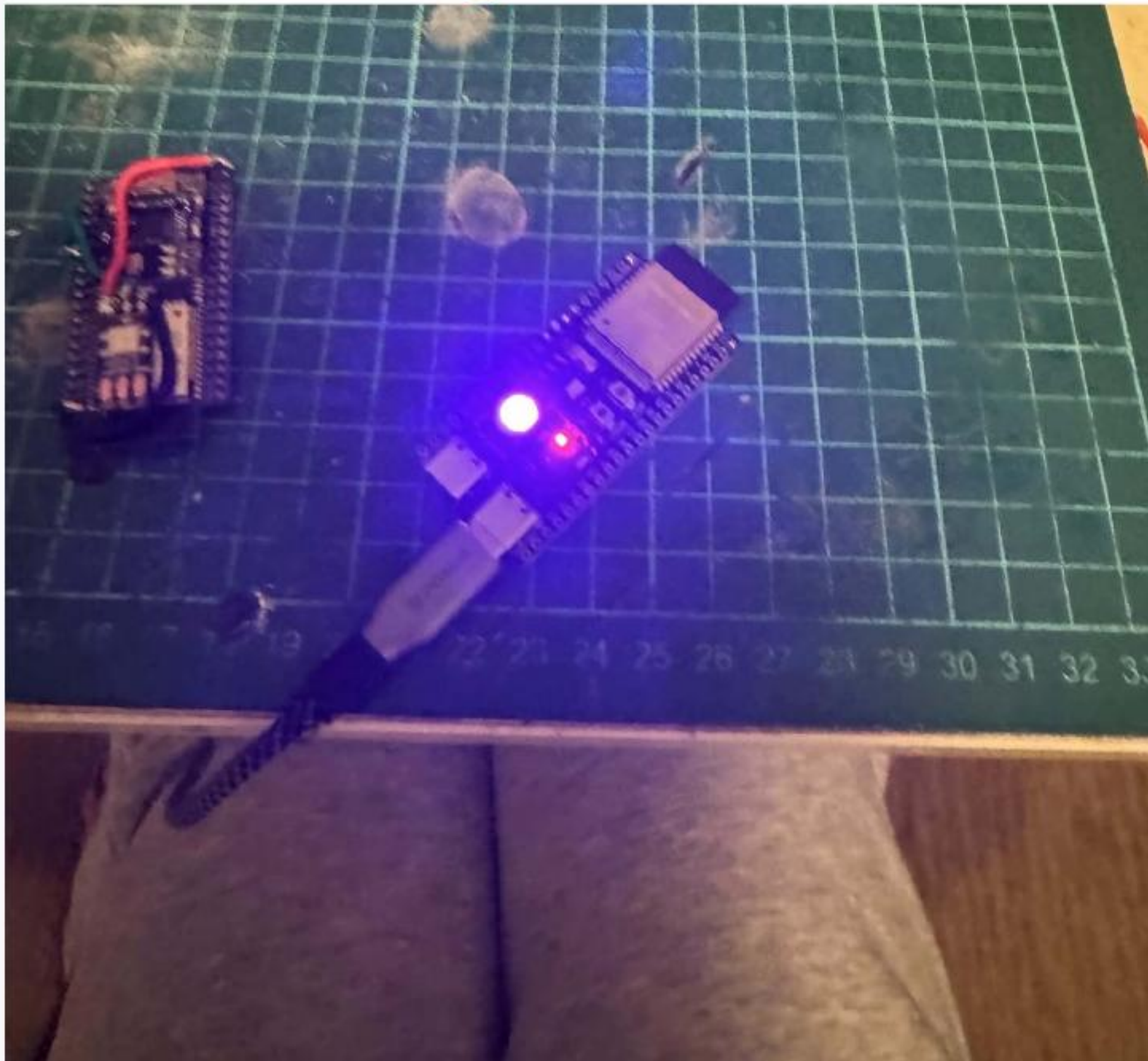
ANNOTATION

- Main operational scanning display on the CYD platform.
- Shows the device in its normal live monitoring interface.
- Use this page as the visual reference for the standard scanning state.

02. ESP32-S3 + ESP32-WROOM - Boot indication (blue)



***ESP32S3 and ESP32 WROOM Boot
light- blue***



ANNOTATION

- Blue is the startup / boot indication.
- The two hardware formats are shown together for direct visual comparison.
- This state appears before normal ready/scanning indication.

03. ESP32-S3 - Bare board



ANNOTATION

- Unassembled ESP32-S3 development board used for the compact build.
- Useful identification image before wiring, enclosure, or final assembly.
- The board is shown from the component side.

04. ESP32-WROOM - Fully assembled with LED



ANNOTATION

- Completed WROOM-based board assembly with status LED and wiring fitted.
- Shows the compact physical arrangement of the finished electronics.
- Use this image as an assembly-orientation reference.

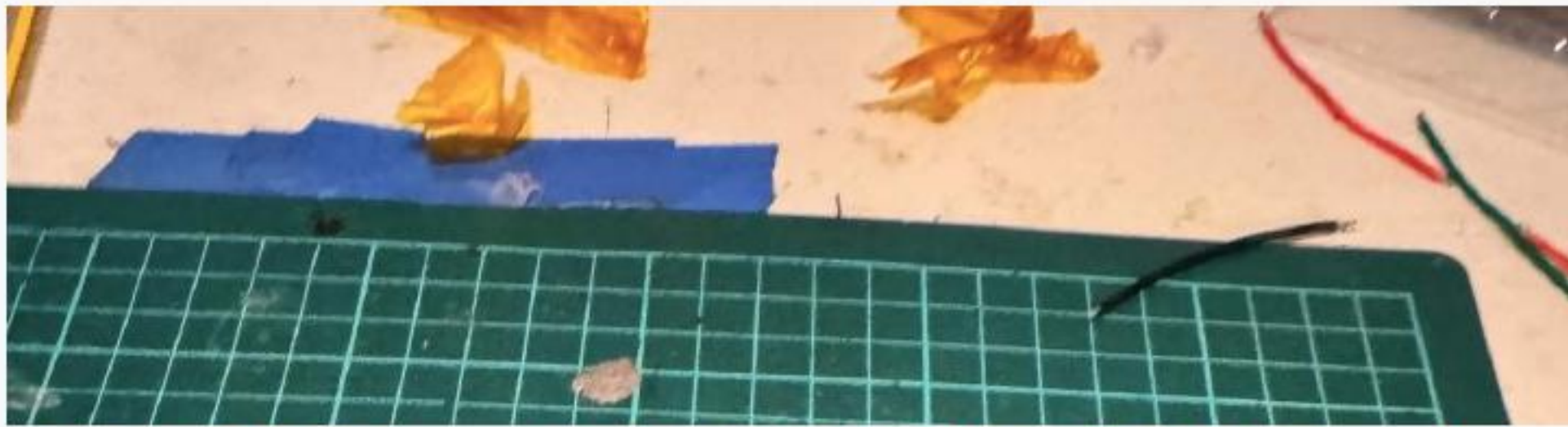
05. CYD / CYD Expanded - Splash page



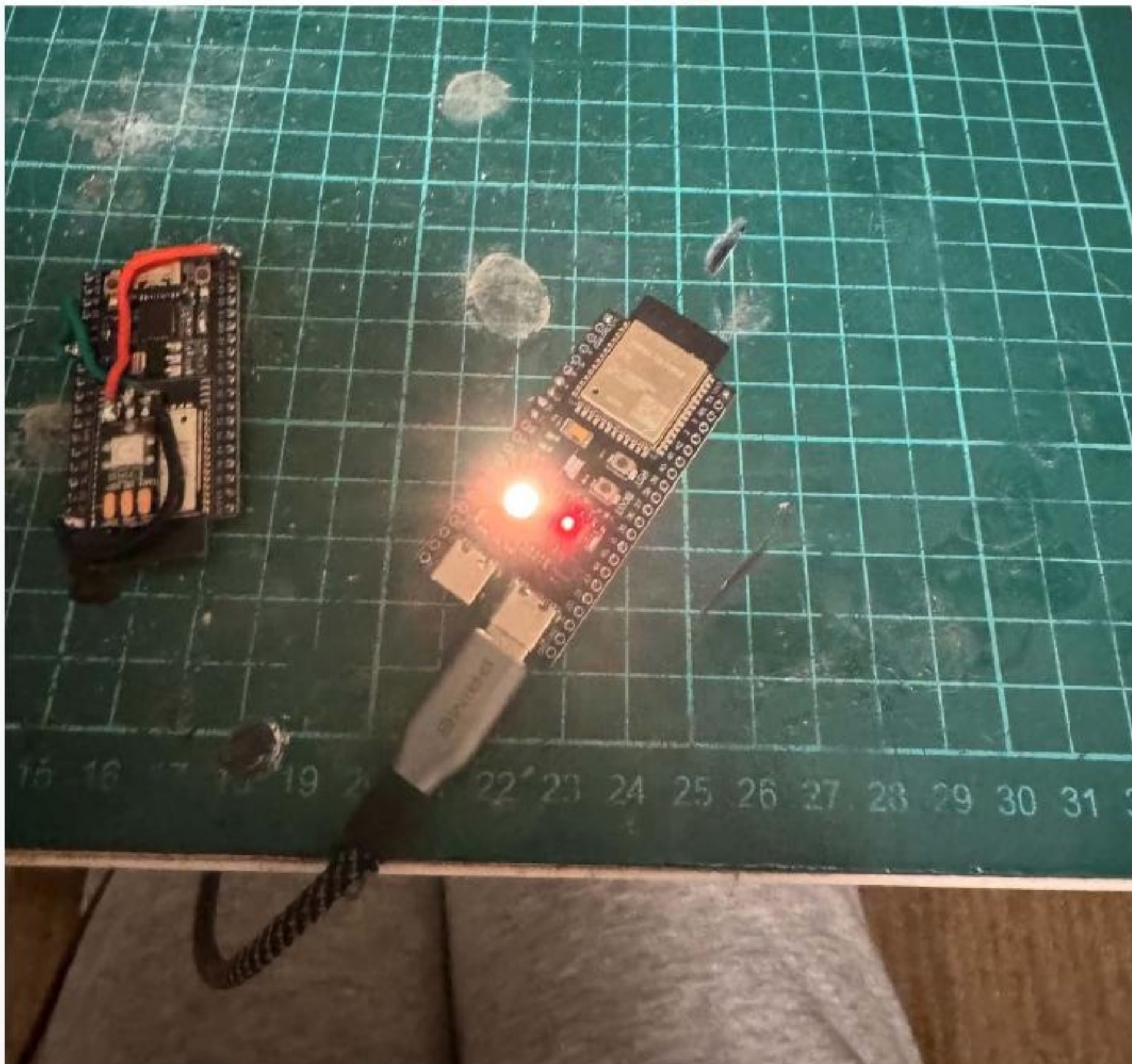
ANNOTATION

- Startup splash/identification screen on the CYD display.
- This appears in the startup flow before the main operating interface.
- Useful for confirming the unit has powered and reached the display stage.

06. ESP32-S3 + ESP32-WROOM - Warning indication (orange)



ESP32S3 and ESP32 WROOM Warning light- Orange



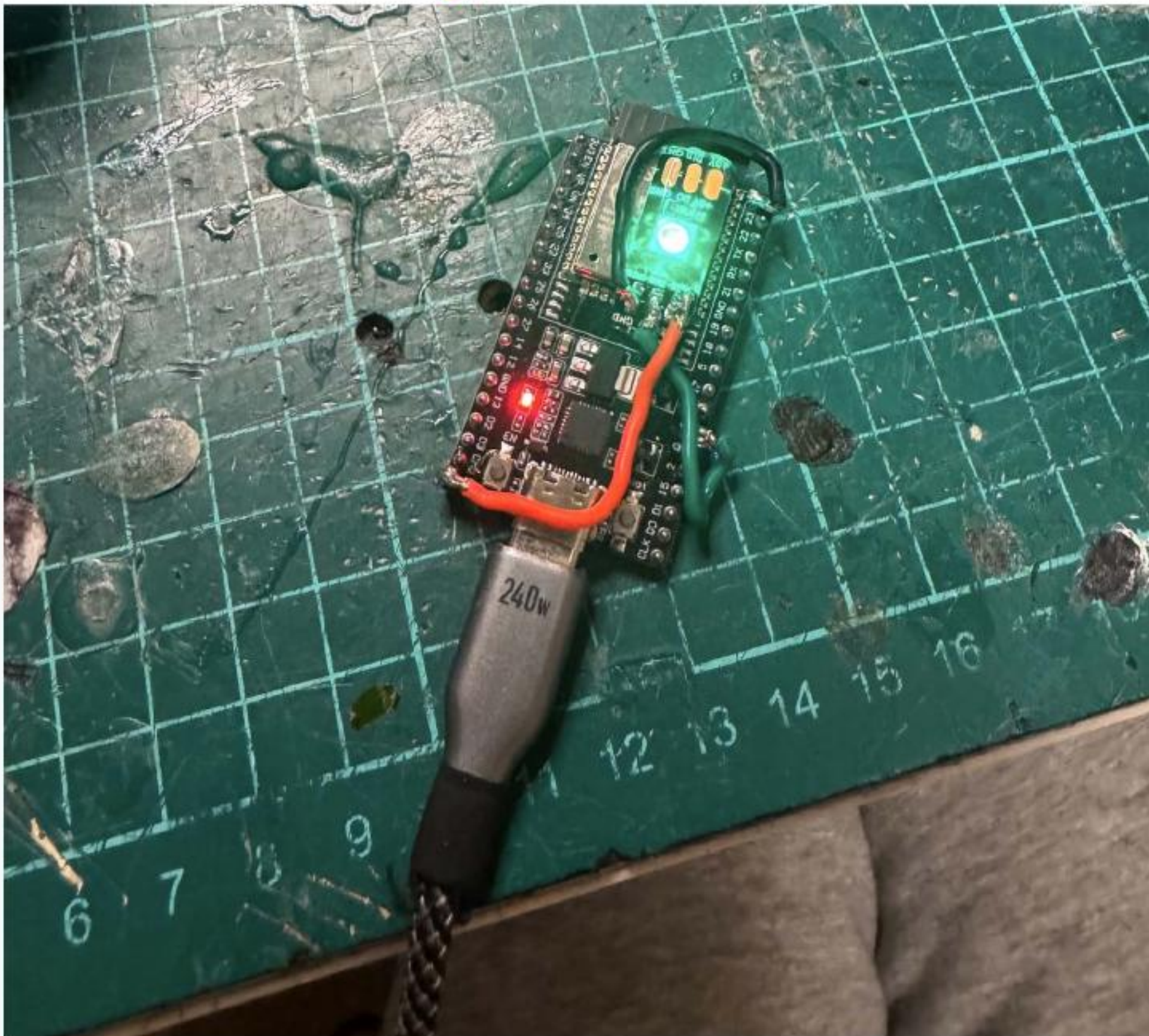
ANNOTATION

- Orange is the warning / possible-detection indication.
- The same warning colour is demonstrated on both hardware builds.
- It provides a visible intermediate warning before the highest alert state.

07. ESP32-S3 + ESP32-WROOM - Status indication (green)



ESP32S3 and ESP32 WROOM Status light- Green



ANNOTATION

- Green is the normal ready / scanning status indication.
- Both builds are shown operating with the same status convention.
- This is the normal state to expect after successful startup.

08. CYD / CYD Expanded - SYS STAT page



ANNOTATION

- System-status and diagnostic screen on the CYD platform.
- Provides runtime information for checking the operating state of the unit.
- Useful during setup, testing, and troubleshooting.

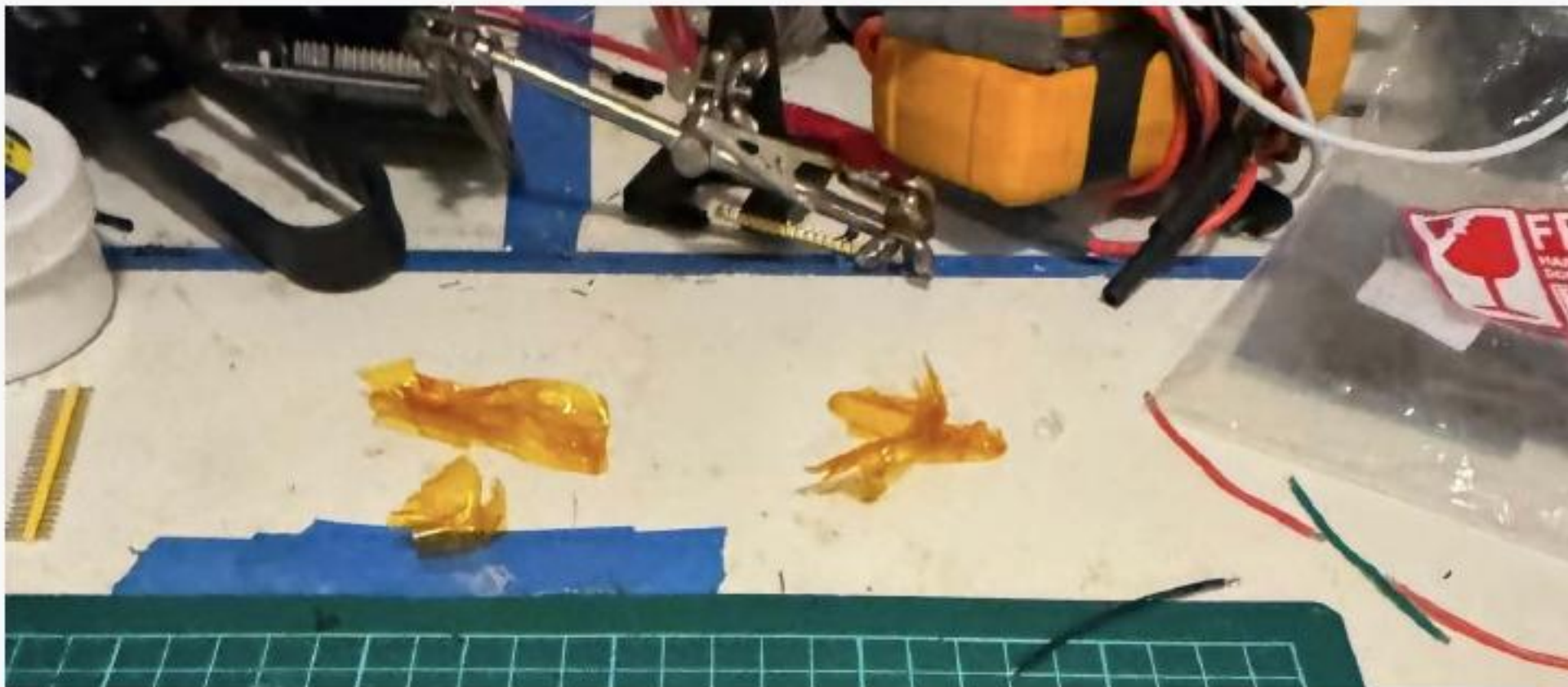
09. ESP32-S3 + ESP32-WROOM - High warning indication (red)



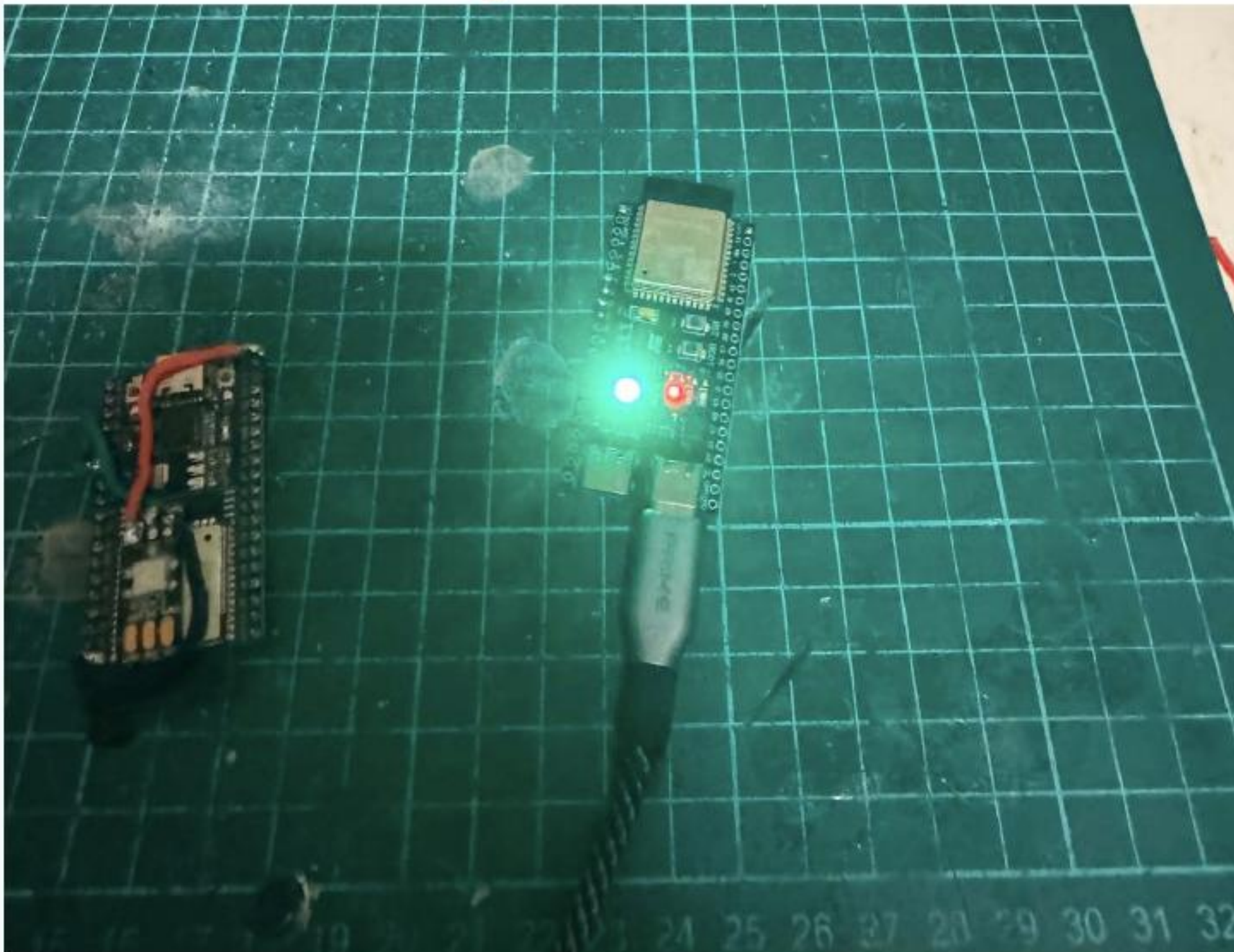
ANNOTATION

- Red is the high-priority / high-confidence alert indication.
- Designed to be immediately visible during operation.
- The photo demonstrates the alert colour on the ESP32-S3 alongside the WROOM build.

10. ESP32-S3 + ESP32-WROOM - Status indication (green)



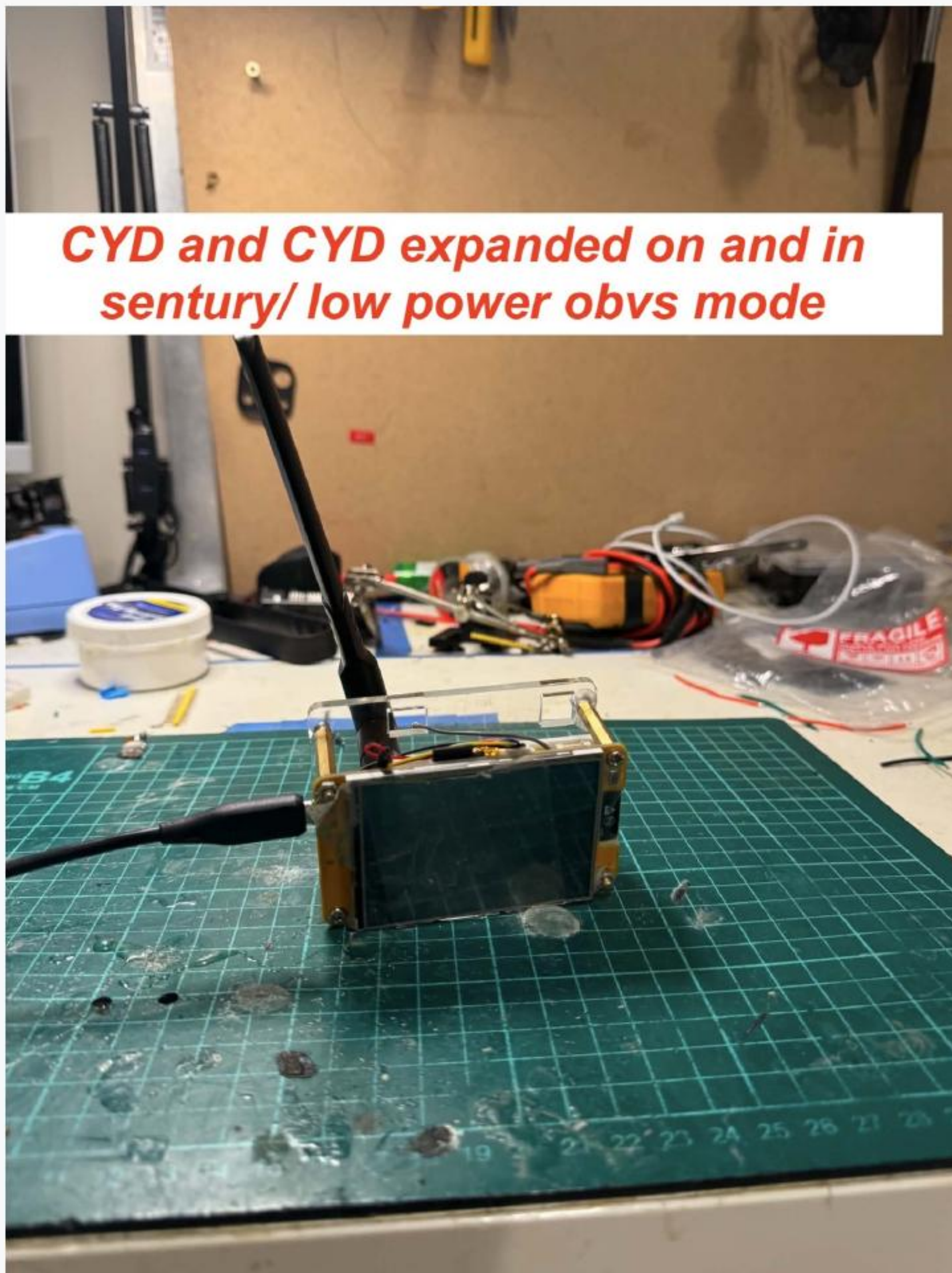
ESP32S3 and ESP32 WROOM Status light- Green



ANNOTATION

- Alternate view of the normal green ready/scanning state.
- Provides a closer visual reference for LED brightness and placement.
- Useful where the side-by-side status image is too wide for detailed inspection.

11. CYD / CYD Expanded - Sentry / low-power mode



ANNOTATION

- Dark-display operating state used for sentry / low-power operation.
- The screen is intentionally unobtrusive while the unit remains in its monitoring mode.
- Use this image to identify the expected appearance of the reduced-visibility state.

12. CYD / CYD Expanded - Legal / privacy page



ANNOTATION

- On-device legal and privacy information page.
- Part of the user-facing startup / acknowledgement flow.
- This page documents the privacy and compliance information presented before normal use.

13. ESP32-WROOM vs ESP32-S3 - Hardware comparison



ANNOTATION

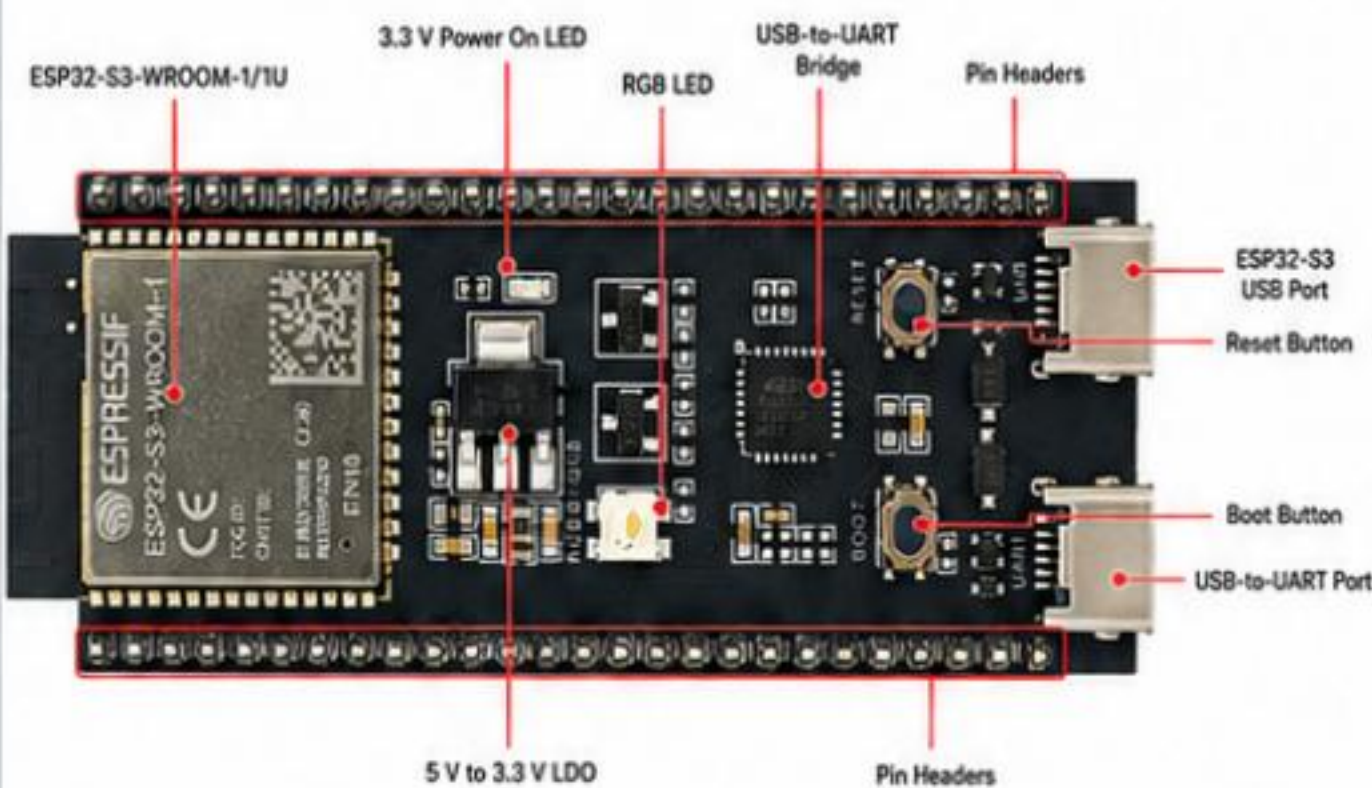
- Side-by-side comparison of the two compact hardware platforms.
- ESP32-WROOM assembly is shown on the left; ESP32-S3 board is shown on the right.
- Useful for identifying the correct board before flashing or assembly.

Quick reference index

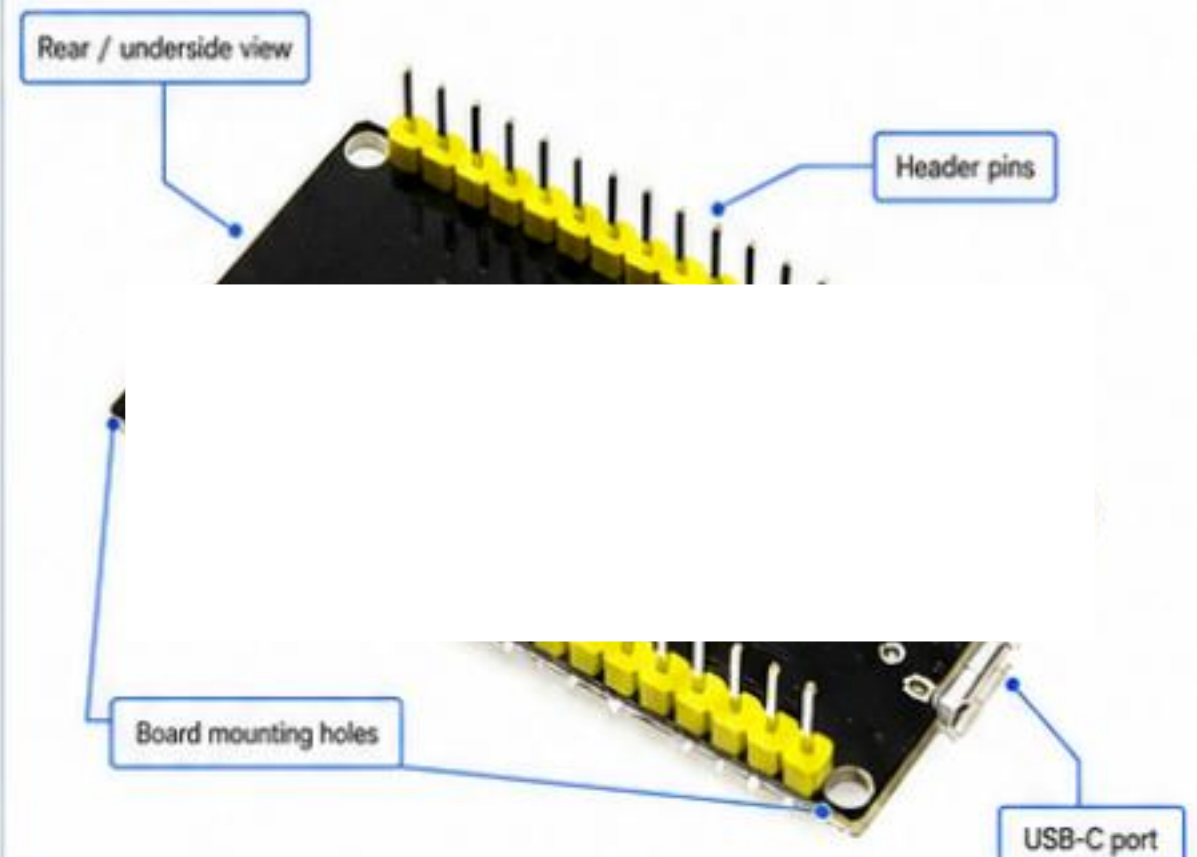
CYD live scanning	Photo 01
Blue boot LED	Photo 02
ESP32-S3 bare board	Photo 03
WROOM assembled board	Photo 04
CYD splash page	Photo 05
Orange warning LED	Photo 06
Green status LED	Photos 07 and 10
CYD SYS STAT	Photo 08
Red high warning LED	Photo 09
CYD sentry / low-power	Photo 11
CYD legal / privacy	Photo 12
WROOM vs S3 comparison	Photo 13

Original photograph labels are retained; added annotations are editorial descriptions for reference use.

Top View (Labeled)



Rear View (Labeled)



About This Board



A compact, efficient ESP32-S3 development board designed for portable detector projects. It mounts behind your phone and connects with a short USB-C cable.

Yellow pin-header strips are optional and are not required for normal detector operation.



Daily portable / travel build



Very little practical skill required



USB-C power and programming



Onboard RGB LED



Approx. 20% extra phone battery over 8 hours

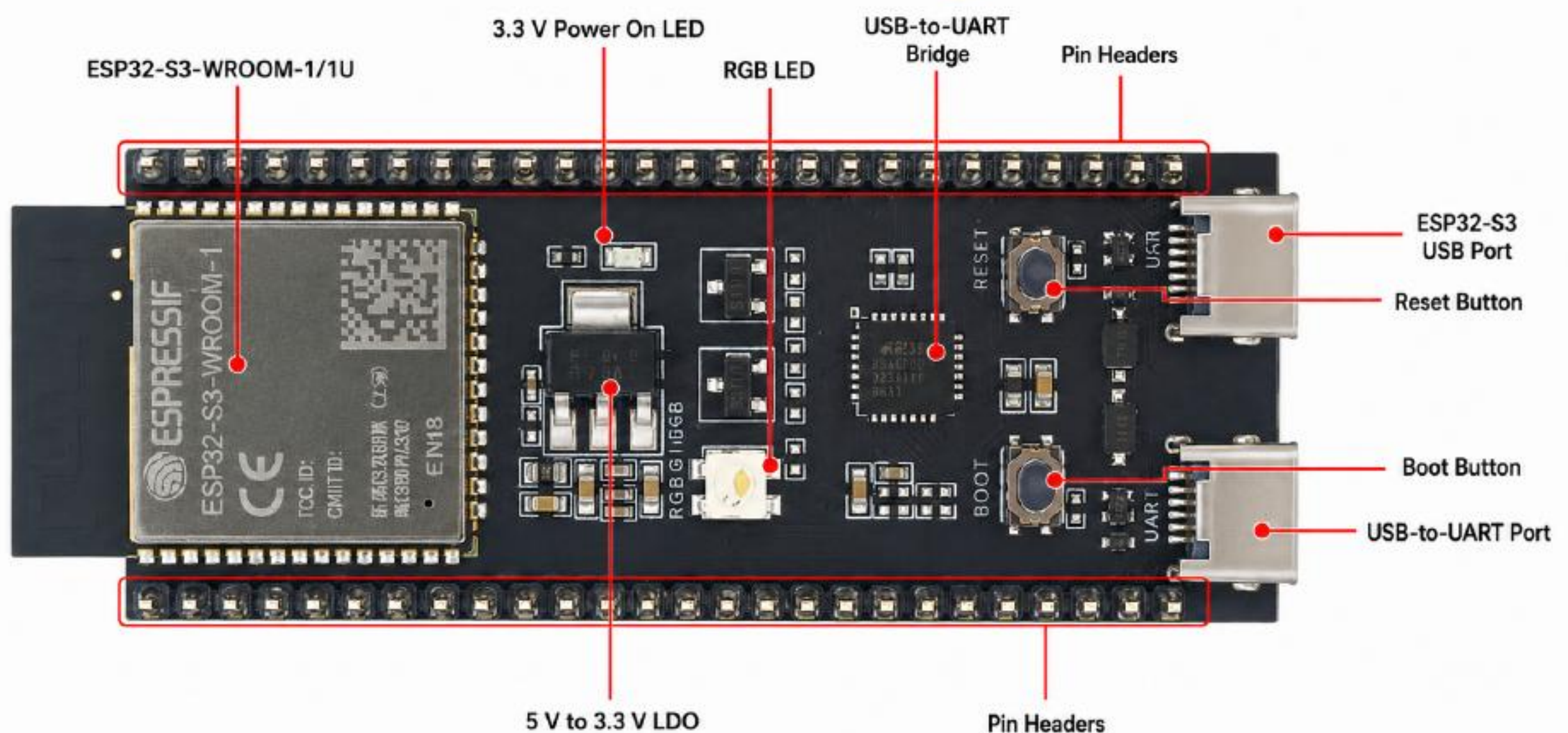


Designed to mount behind a phone with a short USB-C cable

ESP32-S3 Lite - Board Layout

Detailed component and connector identification

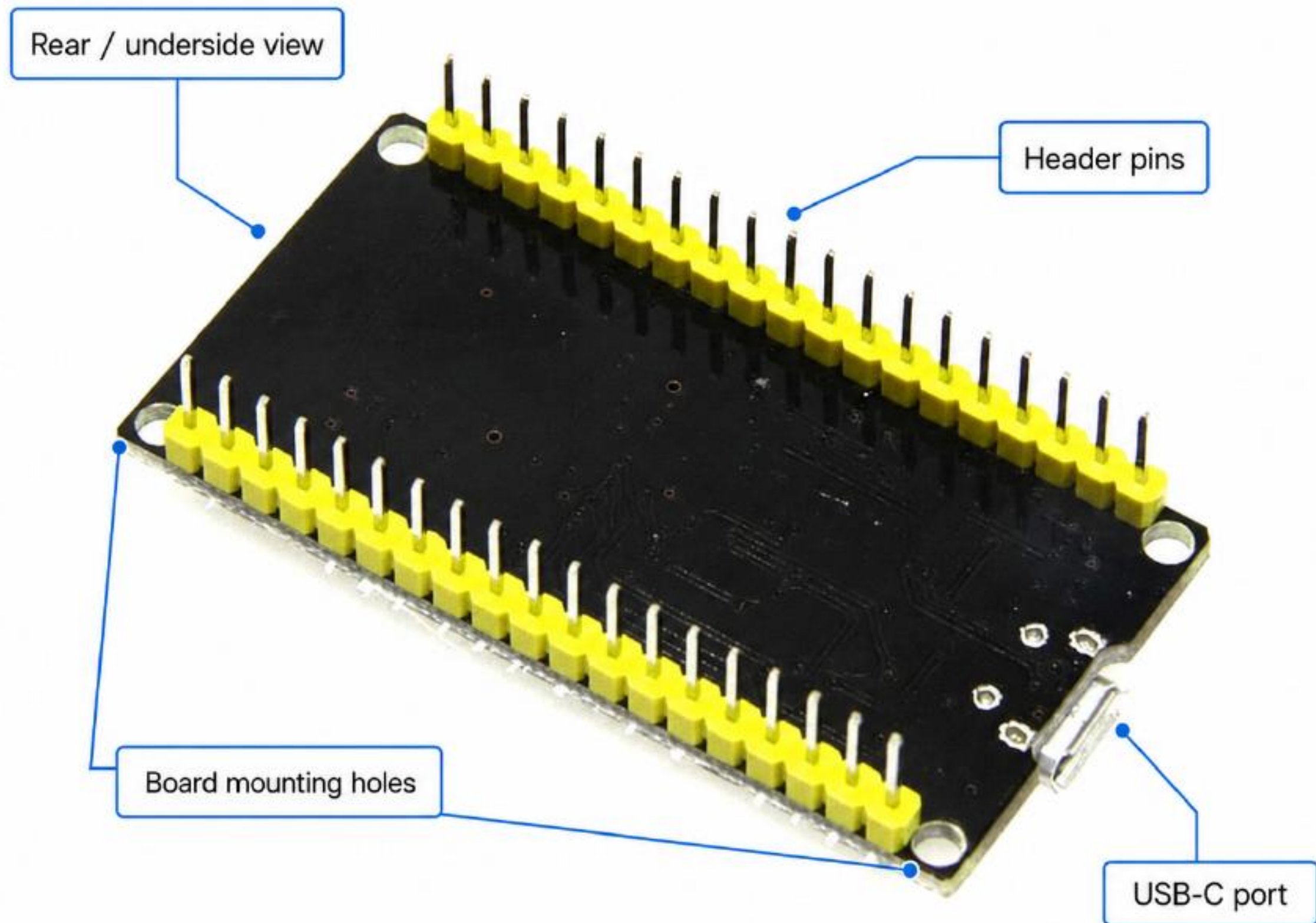
ESP32-S3 Board Layout



ANNOTATION

- Labeled top-view reference for the preferred ESP32-S3 Lite hardware.
- Shows the ESP32-S3-WROOM module, onboard RGB LED, BOOT/RESET controls and the two USB-C interfaces.
- The portable S3 build is intended to use USB-C for power/programming and requires no assembly tools.
- The yellow pin-header strips shown with the board are optional; they do not need to be fitted for normal detector operation.

ESP32-S3 Development Board — Rear View



ANNOTATION

- Rear-view reference showing the underside, header positions, mounting holes and USB-C connector position.
- Useful for checking orientation before fitting the board behind a phone or inside a case.
- The yellow pin-header strips are optional and can be left unfitted for the standard ESP32-S3 Lite detector build.
- Board revisions can differ slightly; confirm the exact board purchased before relying on mechanical dimensions.

ESP32 Wroom - Preferred USB-C hardware

USB-C version preferred over Micro-USB for the portable WROOM build

2. ESP32-WROOM-32 (with USB-C)

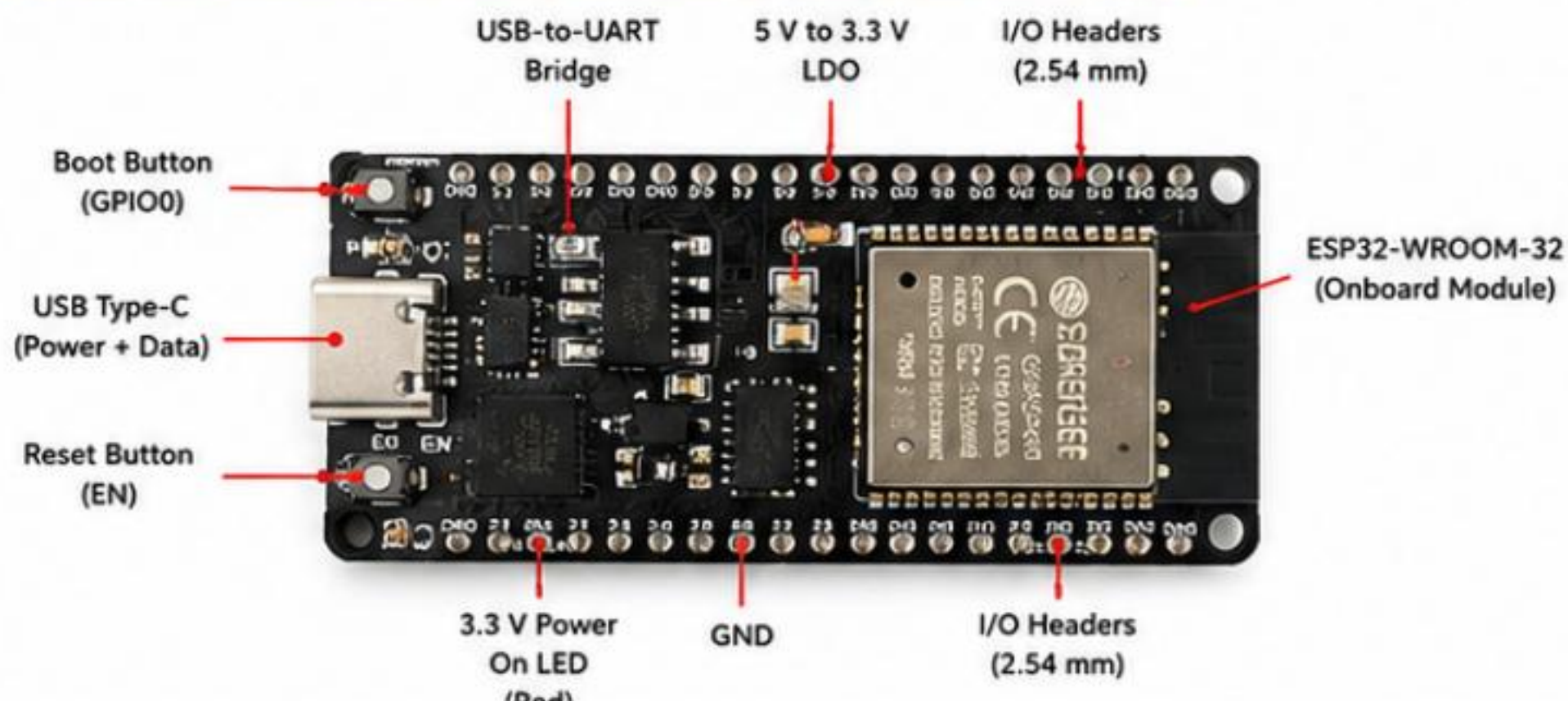


KEY FEATURES

- USB Type-C
- ESP32-WROOM-32 module (onboard)
- 3.3 V logic
- Boot button
- Reset button
- USB-to-UART bridge
- 5 V to 3.3 V LDO

✓ **RECOMMENDED
FOR WROOM BUILD**

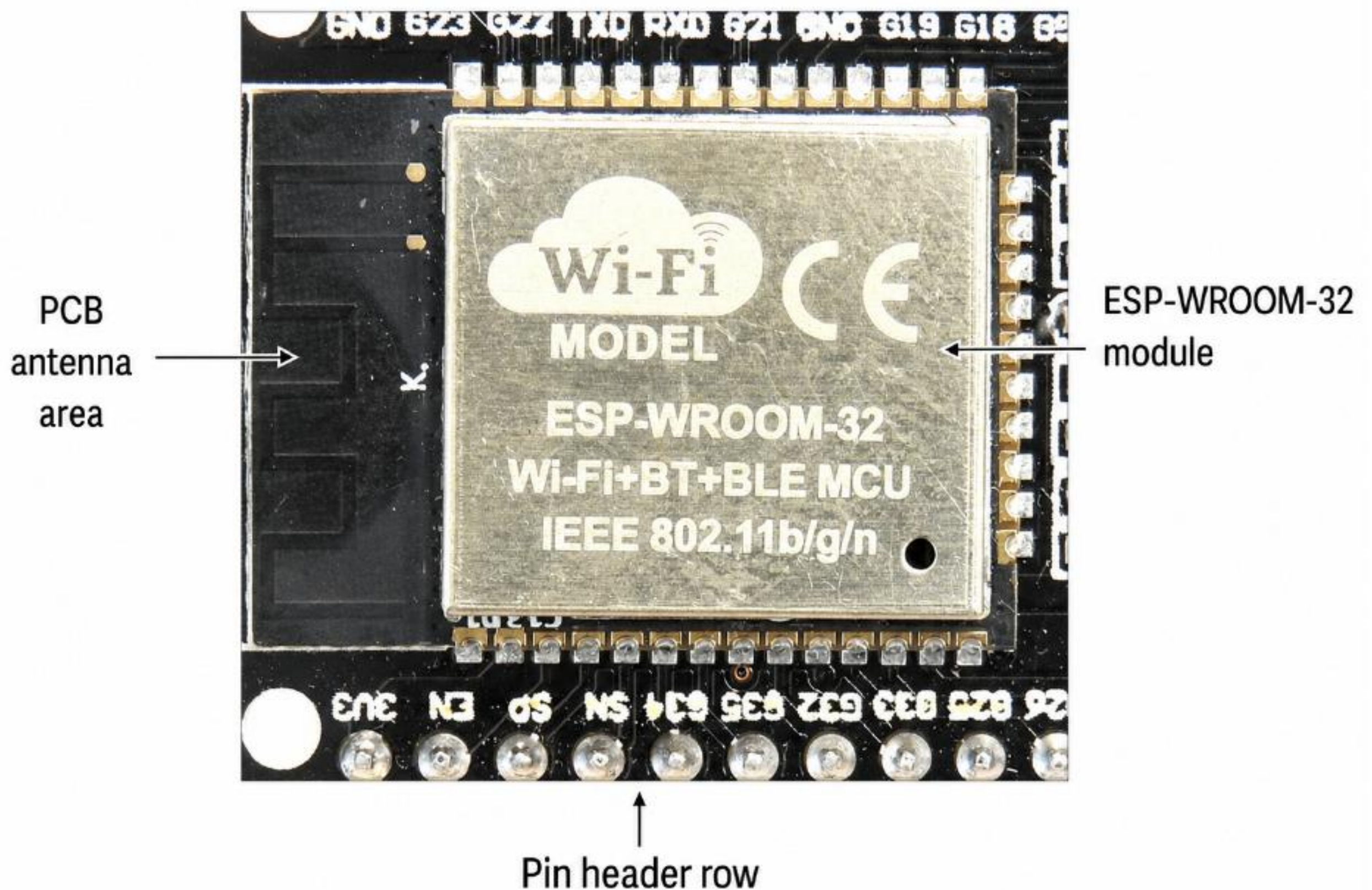
4. ESP32 WROOM-32 PINOUT OVERVIEW (Top View)



ANNOTATION

- Preferred WROOM board uses an ESP-WROOM-32 module with a USB-C power/data connector.
- The project WROOM build uses a separate WS2812B / NeoPixel status LED on GPIO 4.
- USB-C-equipped boards are preferred for easier cable compatibility with the portable phone-powered setup.
- The yellow pin-header strips shown with the board are optional; they are not required for normal detector operation.
- Confirm the actual pinout and USB-to-UART implementation of the board revision purchased.

ESP-WROOM-32 Module Close-Up



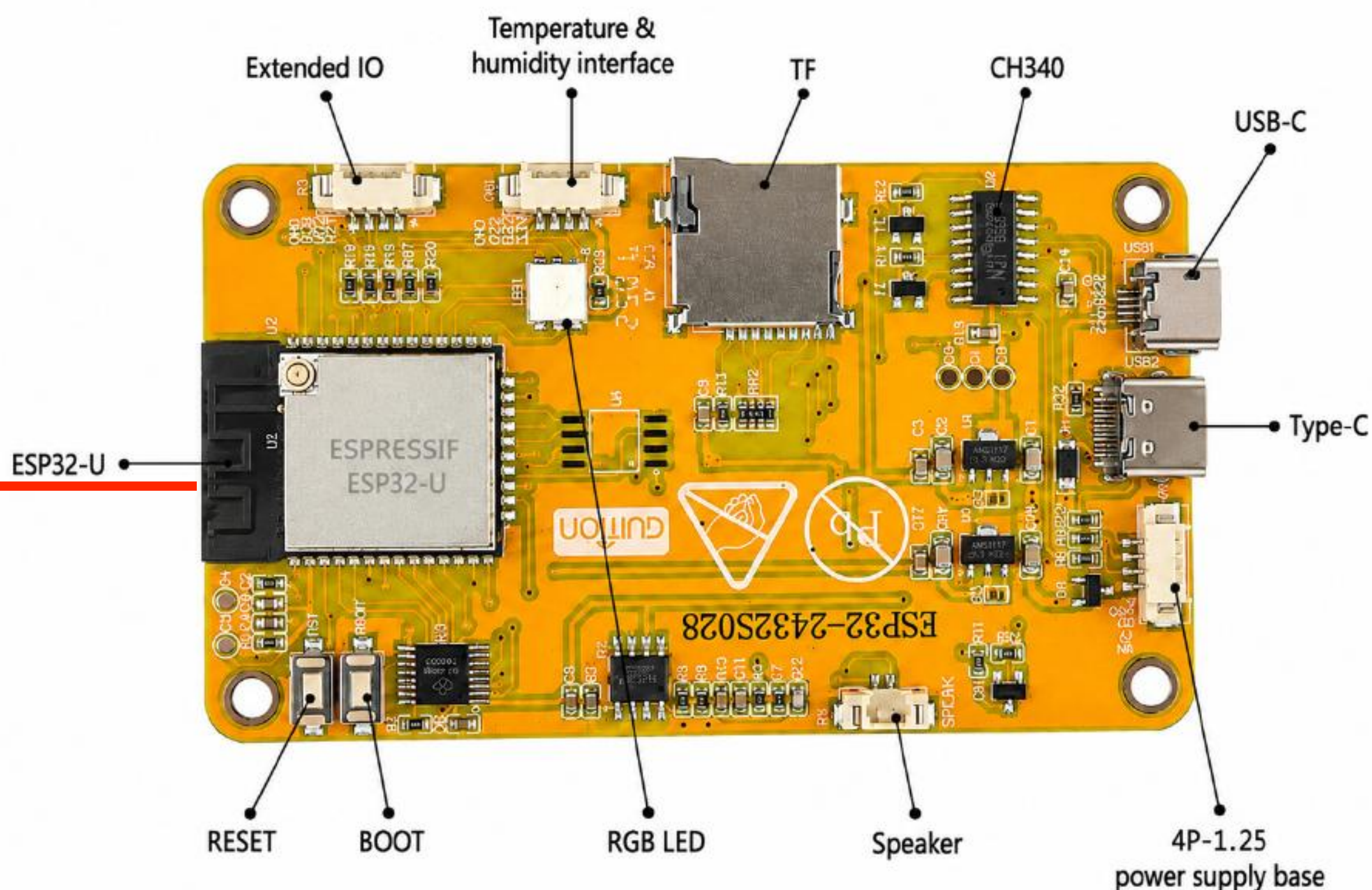
ANNOTATION

- Close-up of the classic ESP-WROOM-32 radio module and its onboard PCB antenna area.
- The portable WROOM build uses the module's onboard antenna rather than the CYD external-antenna arrangement.
- This image is useful when identifying the correct classic ESP32/WROOM hardware before flashing.

CYD DEV Lite / Expanded - Preferred CYD hardware

ESP32-2432S028R CYD - preferred external-antenna-capable revision

ESP32-2432S028R CYD Board Layout



ANNOTATION

- Preferred CYD revision uses an external-antenna-capable ESP32-U onboard radio module.
- USB-C power/programming ports are preferred over Micro-USB when sourcing CYD boards.
- The corrected layout shows the ESP32-U module, U.FL / IPEX connector, TF/SD slot, BOOT/RESET controls and USB-C interfaces.
- For external-antenna operation, the RF selector must be moved to the U.FL / IPEX path before the external antenna is considered active.

CYD DEV Lite - 3 dBi external antenna



ANNOTATION

- CYD DEV Lite uses a 3 dBi 2.4 GHz external antenna in the preferred external-antenna configuration.
- The pictured style uses a U.FL / IPEX pigtail to an external threaded antenna connector and removable whip antenna.
- Use only after confirming the CYD has the correct ESP32-U / U.FL-IPEX hardware and the RF selector has been moved to the external path.
- Actual reception depends on antenna quality, orientation, environment, interference and the transmitting device.

CYD Expanded DEV - 8 dBi external antenna



ANNOTATION

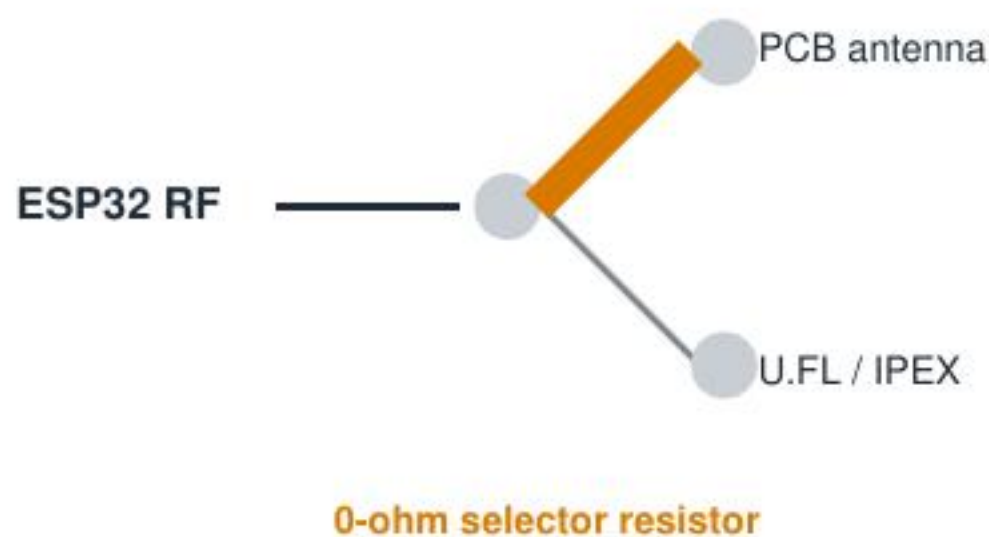
- CYD Expanded DEV uses an 8 dBi 2.4 GHz external antenna in the preferred stationary configuration.
- The higher-gain antenna is intended primarily to improve practical BLE reception for fixed/development use.
- The CYD must have external-antenna-capable hardware and the 0-ohm RF selector must be moved to the U.FL / IPEX path.
- Do not assume that plugging an antenna into the connector automatically selects the external RF path.

CYD external antenna - Required 0-ohm RF selector change

Required for both CYD DEV Lite (3 dBi) and CYD Expanded DEV (8 dBi) external-antenna configurations

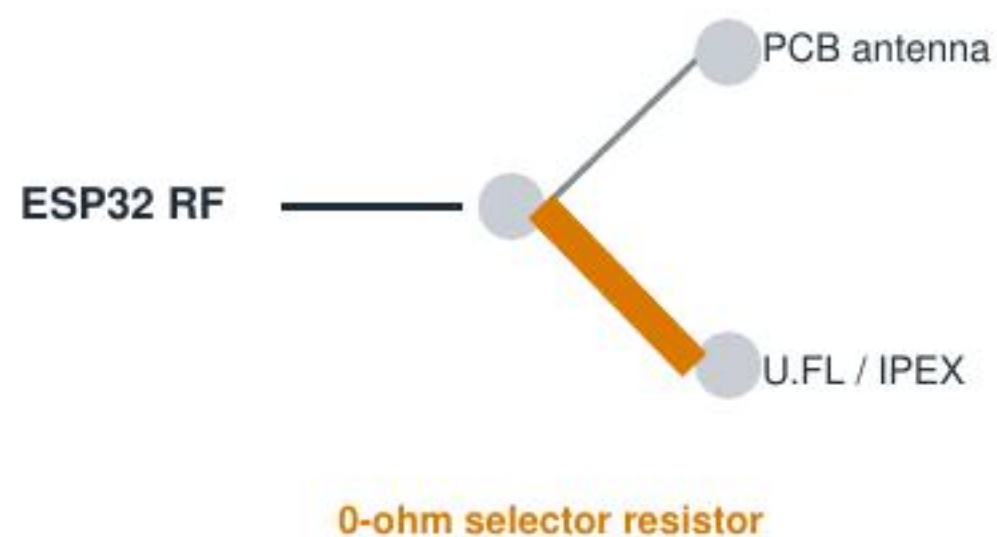
REQUIRED ASSEMBLY STEP Disconnect all power before soldering. The U.FL / IPEX socket is not automatically selected just because it is fitted to the board.

BEFORE - PCB ANTENNA SELECTED



External antenna NOT selected

AFTER - U.FL / IPEX SELECTED



External antenna ACTIVE

REQUIRED PROCEDURE

- 1. Disconnect USB and every other power source from the CYD.
- 2. Identify the ESP32-U module, PCB-antenna route, U.FL / IPEX connector and the tiny 0-ohm RF selector.
- 3. Confirm which selector position currently routes RF to the onboard PCB antenna.
- 4. Remove the 0-ohm resistor from the PCB-antenna position using fine SMD tools and magnification.
- 5. Move/resolder the same resistor so the common RF feed is connected to the U.FL / IPEX path only.
- 6. Do not bridge both antenna paths. Inspect carefully for solder bridges, lifted pads or damage.
- 7. Connect the 3 dBi (CYD DEV Lite) or 8 dBi (CYD Expanded DEV) antenna only after the RF path is correctly selected.
- 8. Board revisions can differ: identify the selector on the exact board received before modifying it.

Expanded quick reference index

Original annotated photo book	Pages 1-16
ESP32-S3 Lite hardware guide	Page 17
S3 Lite board layout	Page 18
S3 Lite rear / underside view	Page 19
ESP32 Wroom preferred USB-C hardware	Page 20
ESP-WROOM-32 radio module close-up	Page 21
CYD preferred ESP32-U / USB-C hardware	Page 22
CYD DEV Lite 3 dBi antenna	Page 23
CYD Expanded DEV 8 dBi antenna	Page 24
Required 0-ohm RF selector change	Page 25
Expanded quick reference index	Page 26

HARDWARE CONFIGURATION CARRIED FORWARD

- ESP32-S3 Lite: USB-C portable board with onboard RGB status LED.
- ESP32 Wroom: USB-C board preferred; separate WS2812B / NeoPixel status LED.
- Yellow pin-header strips shown with the S3 and WROOM boards are optional for normal detector operation.
- CYD DEV Lite: preferred ESP32-U / U.FL-IPEX CYD, USB-C ports preferred, 3 dBi antenna.
- CYD Expanded DEV: preferred ESP32-U / U.FL-IPEX CYD, USB-C ports preferred, 8 dBi antenna.
- External-antenna CYD builds require the 0-ohm RF selector to be moved to U.FL / IPEX.